Integrating from t to $t+h$ and multiplying by $\exp(-\gamma t)$, we arrive at the solution

$$\boldsymbol{p}(t+h) = e^{-\gamma h}\boldsymbol{p}(t) + \sqrt{\mathrm{k_B T}(1-e^{-2\gamma h})}\boldsymbol{M}^{1/2}\mathbf{Z}, \tag{7.3}$$

where $\mathbf{Z}$ is a vector of independent, standard normal random variables. Alternatively, we may write this as a discrete iteration in terms of a random map:

$$\boldsymbol{p}_{n+1} = e^{-\gamma h}\boldsymbol{p}_n + \sqrt{\mathrm{k_B T}(1-e^{-2\gamma h})}\boldsymbol{M}^{1/2}\mathbf{R}_n,$$

with normally distributed $\mathbf{R}$. For later purposes, we will think of this update as acting on both $\boldsymbol{q}$ and $\boldsymbol{p}$, leaving $\boldsymbol{q}$ unchanged, i.e.

$$(\boldsymbol{q}_{n+1}, \boldsymbol{p}_{n+1}) = (\boldsymbol{q}_n, e^{-\gamma h}\boldsymbol{p}_n + \sqrt{\mathrm{k_B T}(1-e^{-2\gamma h})}\boldsymbol{M}^{1/2}\mathbf{R}_n).$$

The Fokker-Planck operator for the OU system (7.2) can be computed directly as

$$\mathcal{L}_{\mathrm{O}}^{\dagger}\phi(\boldsymbol{q},\boldsymbol{p}) = \gamma\nabla_p\cdot(\boldsymbol{p}\phi) + \gamma\mathrm{k_B T}\nabla_p\cdot\boldsymbol{M}\nabla_p\phi.$$

The Ornstein-Uhlenbeck process models the momentum of a Brownian particle, and in deriving it we thought of γ as being associated to the collisional frequency with the particles of the bath. Alternatively, if we ignore the physical origin of this parameter, we can think of γ as an arbitrary coefficient; the sampling of the invariant measure does not depend on this choice. Thus we have a family of *sampling iterations* for the Gaussian distribution, defined for different choices of γ.

7.3.1 Splitting Methods for Langevin Dynamics

We will now consider a family of iterations for molecular systems which are constructed from simple building blocks based on the splitting of an SDE, specifically Langevin dynamics:

$$\begin{aligned} \mathrm{d}\boldsymbol{q} &= \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \\ \mathrm{d}\boldsymbol{p} &= -\nabla U(\boldsymbol{q})\mathrm{d}t - \gamma\boldsymbol{p}\mathrm{d}t + \sqrt{2\gamma\mathrm{k_B T}}\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}. \end{aligned} \tag{7.4}$$

Just as in the deterministic setting (for Newton's equations of motion) we construct such splitting methods by way of an additive decomposition of the vector field, where the differential equations corresponding to any individual piece can be solved exactly. Methods are built from a sequence of updates corresponding to an exact solve of each piece. By "exact solve" we here mean the construction of a method of exactly sampling the distribution generated by the corresponding component. The